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C:\Users\Sai Saahas> mongosh mongodb+srv://<credentials>@cluster0.hqljv0w.mongodb.net/
Microsoft Windows [Version 10.0.26200.9168]
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C:\Windows\System32> cd C:\Users\Sai Saahas\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin

C:\Users\Sai Saahas\Downloads\mongosh-2.10.0-win32-x64\mongosh-2.10.0-win32-x64\bin> mongosh "mongodb+srv://cluster0.hqljv0w.mongodb.net/" --apiVersion 1 --username saahasvaddi_db_user --password sZlrFsADvS4rhhte
Current Mongosh Log ID: 6a8676a13844f6467d5e07c6
Connecting to:      mongodb+srv://<credentials>@cluster0.hqljv0w.mongodb.net/?appName=mongosh+2.10.0
Using MongoDB:      8.0.29 (API Version 1)
Using Mongosh:      2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

Atlas atlas-gubp69-shard-0 [primary] test> use schoolDB;
switched to db schoolDB
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.insertOne({name: "Alice", age: 21, major: "Computer Science" })
{
  acknowledged: true,
  insertedId: ObjectId('6a8679a13844f6467d5e07c7')
}
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.insertMany([
  { name: "Charlie", age: 23, major: "Physics" },
  { name: "David", age: 21, major: "Computer Science" },
  { name: "Eve", age: 22, major: "Mathematics" }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a867b2e3844f6467d5e07c8'),
    '1': ObjectId('6a867b2e3844f6467d5e07c9'),
    '2': ObjectId('6a867b2e3844f6467d5e07ca'),
    '3': ObjectId('6a867b2e3844f6467d5e07cb')
  }
}
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.find({ major: "Computer Science" })
[
  {
    _id: ObjectId('6a8679a13844f6467d5e07c7'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07ca'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  }
]

```

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Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.find({ major: "Computer Science" })
[
  {
    _id: ObjectId('6a8679a13844f6467d5e07c7'),
    name: 'Alice',
    age: 21,
    major: 'Computer Science'
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07ca'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  }
]
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.updateOne({ name: "Alice" }, { $set: { age: 22 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.deleteOne({ name: "Alice" })
{ acknowledged: true, deletedCount: 1 }
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.createIndex({ major: 1 });
major_1
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.getIndexes();
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { major: 1 }, name: 'major_1' }
]
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.createIndex({major: 1, age: -1});
major_1_age_-1
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.dropIndex({"major_1"});
Uncaught:
SyntaxError: Unexpected token (1:32)
> 1 | db.students.dropIndex({"major_1"});

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[
  { _id: 'Computer Science', total: 1 },
  { _id: 'Physics', total: 1 },
  { _id: 'Mathematics', total: 2 }
]
```

```
[
  { _id: 'Computer Science', avgAge: 21 },
  { _id: 'Physics', avgAge: 23 },
  { _id: 'Mathematics', avgAge: 22 }
]
```

```
[
  {
    _id: ObjectId('6a867b2e3844f6467d5e07c9'),
    name: 'Charlie',
    age: 23,
    major: 'Physics'
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07c8'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07cb'),
    name: 'Eve',
    age: 22,
    major: 'Mathematics'
  },
]
```

```

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[
  {
    _id: ObjectId('6a867b2e3844f6467d5e07c9'),
    name: 'Charlie',
    age: 23,
    major: 'Physics'
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07c8'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07cb'),
    name: 'Eve',
    age: 22,
    major: 'Mathematics'
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07ca'),
    name: 'David',
    age: 21,
    major: 'Computer Science'
  }
]
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.courses.insertMany([
  { course: "DBMS", major: "Computer Science" },
  { course: "Algebra", major: "Mathematics" },
  { course: "Quantum Physics", major: "Physics" }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a867f053844f6467d5e07cc'),
    '1': ObjectId('6a867f053844f6467d5e07cd'),
    '2': ObjectId('6a867f053844f6467d5e07ce')
  }
}
Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.aggregate([
  { $lookup: {
    from: "courses",
    localField: "major",

```

```

Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.courses.insertMany([
  { course: "DBMS", major: "Computer Science" },
  { course: "Algebra", major: "Mathematics" },
  { course: "Quantum Physics", major: "Physics" }
]);

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a867f053844f6467d5e07cc'),
    '1': ObjectId('6a867f053844f6467d5e07cd'),
    '2': ObjectId('6a867f053844f6467d5e07ce')
  }
}

Atlas atlas-gubp69-shard-0 [primary] schoolDB> db.students.aggregate([
  { $lookup: {
    from: "courses",
    localField: "major",
    foreignField: "major",
    as: "enrolledCourses"
  }}
]);

[
  {
    _id: ObjectId('6a867b2e3844f6467d5e07c8'),
    name: 'Bob',
    age: 22,
    major: 'Mathematics',
    enrolledCourses: [
      {
        _id: ObjectId('6a867f053844f6467d5e07cd'),
        course: 'Algebra',
        major: 'Mathematics'
      }
    ]
  },
  {
    _id: ObjectId('6a867b2e3844f6467d5e07c9'),
    name: 'Charlie',
    age: 23,
    major: 'Physics',
    enrolledCourses: [
      {

```

mongosh mongodb+srv://<credentials>@cluster0.hqjiv0w.mongodb.net/

```
{
  _id: ObjectId('6a867b2e3844f6467d5e07c9'),
  name: 'Charlie',
  age: 23,
  major: 'Physics',
  enrolledCourses: [
    {
      _id: ObjectId('6a867f053844f6467d5e07ce'),
      course: 'Quantum Physics',
      major: 'Physics'
    }
  ]
},
{
  _id: ObjectId('6a867b2e3844f6467d5e07ca'),
  name: 'David',
  age: 21,
  major: 'Computer Science',
  enrolledCourses: [
    {
      _id: ObjectId('6a867f053844f6467d5e07cc'),
      course: 'DBMS',
      major: 'Computer Science'
    }
  ]
},
{
  _id: ObjectId('6a867b2e3844f6467d5e07cb'),
  name: 'Eve',
  age: 22,
  major: 'Mathematics',
  enrolledCourses: [
    {
      _id: ObjectId('6a867f053844f6467d5e07cd'),
      course: 'Algebra',
      major: 'Mathematics'
    }
  ]
}
]
```

Atlas atlas-gubp69-shard-0 [primary] schoolDB>